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Seeing Points Doing It: Pixel-Point State for Calibration-Free Real-World Reinforcement Learning

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Abstract— Real-world reinforcement learning (RL) can learn contact-rich manipulation directly from interaction, but deployment requires perception and rewards without task-specific engineering. We present a calibration-free framework that uses online RGB point tracking as a unified pixel-space interface. Tracked keypoints from uncalibrated views define the policy state and a final-state relational reward. Relations are computed within each view without cross-view association, so neither robot–camera extrinsics nor 3D reconstruction is required. A belief-gated dual-head Soft Actor-Critic policy combines a temporal prior with reward-change evidence through a two-state Bayesian filter. On seven real-world tasks spanning rigid, articulated, and deformable objects, the approach achieves 84.4 % success averaged over tasks and 3 random seeds, outperforming behavior cloning trained on 50 demonstrations per task, vision-based RL, and single-head SAC. Tracking recovers at 92.9 % after 30 s of full-view occlusion with hidden object displacement, and injected keypoint jitter and drift degrade success gradually. The relational success criterion agrees with physical success on 94.3 % of evaluation episodes (Cohen's $\kappa = 0.88$). In a targeted simulation ablation over 10 seeds, Bayesian belief gating reaches 60.8 % success, versus 41.1 % for a capacity-matched single head, 47.2 % for an EMA gate, and 44.5 % for a learned gate, and is the only variant with no collapsed seed. Against ground-truth contact the gate reaches F1 0.87, where fixed, smooth, EMA, learned, and instantaneous gates reach 0.61–0.75, and control success tracks that accuracy: Bayesian temporal inference is an essential component that works by inferring contact, not by adding capacity or smoothing. A separate contact-regime study identifies a sustained stick-slip setting where gating underperforms single-head SAC. Camera-displacement studies support calibration-free deployment up to 20 cm; the relational reward is scoped to static geometric goals.

I. Introduction

Many everyday manipulation tasks depend on sustained physical interaction, where small changes in friction, compliance, or contact mode flip the outcome: objects slide, stick, pivot, or intermittently lose contact, as when pushing fragments into alignment or straightening deformable materials. Analytic models are brittle in this regime and kinematic planning is insufficient, which motivates controllers that adapt online through feedback.

Two dominant paradigms struggle here. Imitation learning reproduces demonstrated behavior efficiently, but demonstrations rarely span the diversity of contact conditions, so policies fail to recover once contact leaves the demonstrated manifold. Sim-to-real transfer provides scalable exploration, yet contact, deformation, and friction are hard to simulate, and modest modeling errors induce brittle real-world behavior; real-to-sim-to-real pipelines narrow this gap but still require task-specific digital twins and privileged training signals [1]. Real-world reinforcement learning (RWRL) discovers recovery behaviors directly through online interaction [2], but two practical problems bottleneck deployment. The first is state: many systems depend on calibrated 3D perception, robot–camera extrinsics, object pose estimation, or physical markers [3], all costly to reproduce and fragile under sensor drift and object diversity. The second is reward: hand-crafted dense shaping is task-specific, brittle, and prone to inducing unintended behavior.

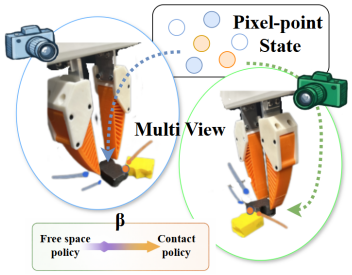


Fig. 1: Calibration-free pixel-point tracking as the state interface for real-world RL. Keypoints tracked in each uncalibrated RGB view define both the policy state and, through their geometric relations to an annotated final-state target, the reward.

We address both with a single interface (Figures 1 and 2). The agent observes a small set of keypoints tracked in multiple uncalibrated RGB views, and the reward is defined by geometric relations among those same keypoints relative to an annotated final state. Each view contributes only within-view geometric relations, and the reward is a sum over views; the system never performs cross-view geometric association or fusion. Cross-view matching is precisely what would require calibrated extrinsics, so avoiding it is what makes the interface calibration-free. This also explains what the interface does not provide: metric 3D quantities are never recovered, and the goal is specified in the image plane.

Physical interaction additionally exhibits distinct dynamics in free space and in contact, and a single policy faces a trade-off between behavior aggressive enough to make progress through contact and behavior conservative enough to be safe outside it. We therefore parameterize the actor with two heads and interpolate them using a scalar belief maintained by a two-state Bayesian filter that combines a temporal prior with reward-change evidence. The targeted ablation in Table VI separates

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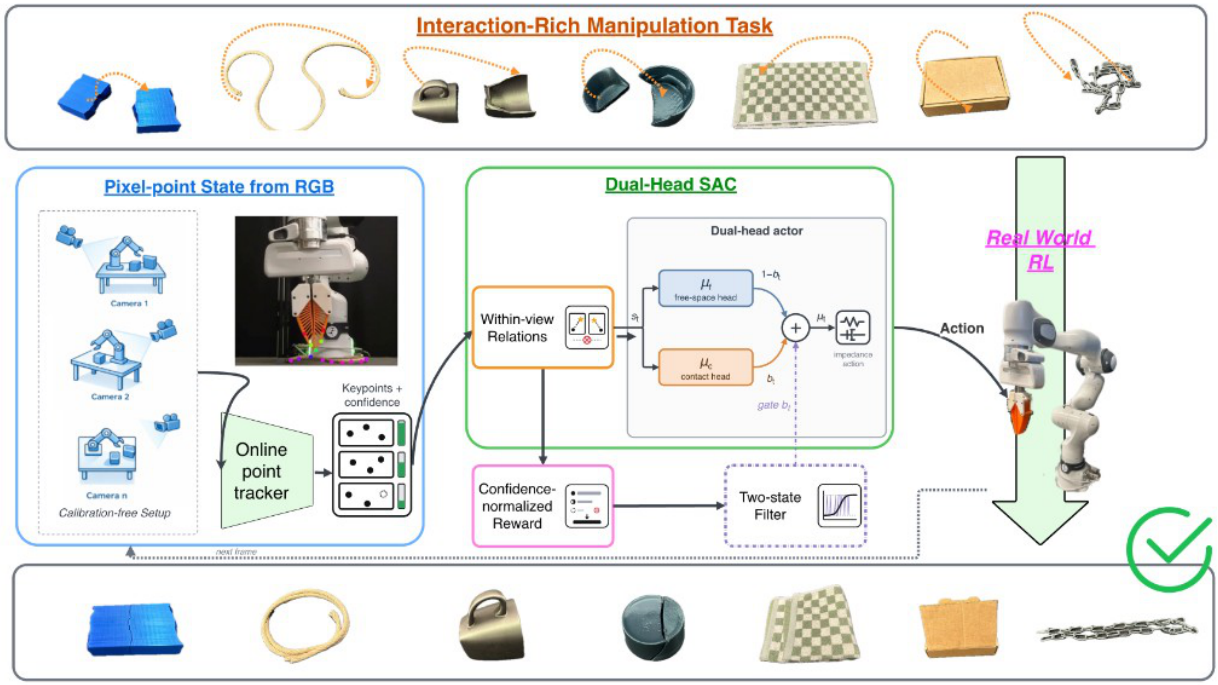


Fig. 2: System overview. An online tracker maps uncalibrated multi-view RGB to per-view keypoints and confidences. Within-view relations define a confidence-normalized reward; a two-state Bayesian filter integrates reward-change evidence over time to gate a dual-head SAC actor (Sections III-C and VI).

this inference from increased actor capacity and generic smoothing: Bayesian belief gating reaches 60.8 % success over 10 seeds, compared with 41.1 % for a capacity-matched single head and 47.2 % for an EMA gate, with no collapsed seed. Against ground-truth contact the Bayesian gate reaches F1 0.87 where the alternatives reach 0.61–0.75. Bayesian temporal inference is thus an essential component in this setting, and its value comes from inferring the contact mode. A separate sustained-contact study characterizes the regime dependence of this benefit (Section VI).

Section II positions this against tracked-point imitation learning and uncertainty-aware control, and Section IV reports the full evaluation protocol. Our contributions are:

- A calibration-free pixel-point state interface for real-world RL that requires no extrinsic calibration, 3D reconstruction, or object pose estimation, because it performs no cross-view geometric association.
- A final-state relational reward defined over tracked keypoints, stated in a confidence-normalized form that removes the degenerate low-confidence optimum admitted by confidence-scaled weighting.
- A real-robot evaluation on seven tasks spanning rigid, articulated, and deformable objects, with results averaged over 3 random seeds and an explicit protocol: interaction budget, 50 demonstrations per task for BC, the single component each baseline changes, and train/test separation, together with a validation of the relational success criterion against physical success (94.3 % agreement, $\kappa = 0.88$) and a characterization of the interface under sustained full occlusion and injected keypoint jitter and drift.
- A controlled gate ablation supporting Bayesian temporal inference as an essential algorithmic component beyond added capacity, fixed blending, generic smoothing, and a learned gate; a ground-truth contact study showing that its control benefit tracks its contact-identification accuracy; and a contact-regime study bounding the scope of this benefit.

II. Related Work

a) Tracked points as a manipulation interface: Tracked 2D points have become a popular intermediate representation for manipulation. ATM [4] pre-trains a model that predicts future tracks of arbitrary points and learns policies from them; Track2Act [5] predicts goal-conditioned tracks from internet video and converts them to actions through a rigid transform plus a residual policy; Point Policy [6] unifies observation and action spaces around semantic key points for imitation from human video; and ATK [7] selects a minimal task-relevant keypoint set for robustness. These systems use tracked points to specify what motion to imitate. We instead use them as the state and reward interface for online reinforcement learning: the representation must support exploration and credit assignment through contact, and the reward must be computable at every control step from the same signal. Track2Act's rigid transform additionally requires calibrated geometry, which our

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within-view formulation avoids. This line inherits from keypoint representations for manipulation—kPAM [8] replacing 6-DoF pose with semantic keypoints, Transporter Networks [9] inferring rearrangement from image features without object pose, and KETO [10] learning task-specific keypoints—as well as dense visual descriptors [11], which typically still assume calibrated geometry or object-specific representation learning.

b) Real-world reinforcement learning: Online RL on physical robots is constrained by data cost and system engineering. Early work learned visuomotor policies end-to-end on real hardware [12], and large-scale off-policy training reached strong grasping performance given abundant autonomous interaction [13]. Recent systems target usability: SERL [14] provides a sample-efficient stack and shows that reward computation and reset mechanisms matter as much as the algorithm, and human-in-the-loop training [15] reduces sample complexity with corrective interventions. Real-to-sim-to-real pipelines [1] avoid unsafe exploration by fine-tuning in task-specific digital twins. Across these systems, state is typically obtained through calibrated sensing or task-specific perception, which is the dependency we remove.

c) Reward specification and mode-dependent control: Avoiding hand-designed shaping is a long-standing goal. VICE [16] learns a classifier reward from goal-state examples, and preference-based methods learn rewards from human comparisons. Our relational reward is deliberately simpler: one annotated final state per view and no learned reward model, at the cost of being restricted to static geometric goals. For mode-dependent behavior, GUAPO [17] switches between a model-based controller and a local RL policy according to pose estimator uncertainty—structurally the closest analogue to our gate, though its uncertainty comes from a calibrated pose estimator.

d) Point tracking: The track-any-point line [18] provides markerless 2D correspondence without fiducials or motion capture [3]. Most are formulated for offline video, but a control loop cannot wait for a whole sequence. We use TAPTRv3 [19], a detection-style tracker, because it performs sequential per-frame updates with visibility prediction and can run streaming inside the loop.

III. Method

A. Calibration-Free Pixel State via Online Point Tracking

We track keypoints with TAPTRv3 [19], which casts point tracking as a query-based detection problem. We select it for a specific reason: unlike trackers formulated for offline video, TAPTRv3 is an online tracker that performs sequential frame-by-frame updates, predicting each point's location and visibility on each newly arriving frame while carrying its query state forward. We maintain that internal state across frames and run it in a streaming manner, which is what makes it usable inside a closed control loop.

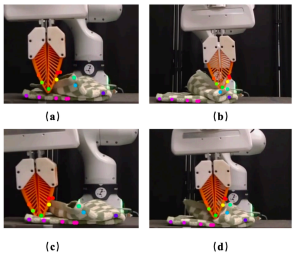


Fig. 3: The interface under deformation. Tracked keypoints stay attached to towel corners and edges through folding and partial occlusion, which is what lets the same relational reward describe deformable and rigid tasks without a per-task perception module.

Figure 2 summarizes the system. Let $t \in \mathbb{N}$ index control steps, $v \in \{1, \dots, V\}$ index RGB cameras, and $k \in \{1, \dots, K\}$ index keypoints. The tracker outputs pixel coordinates and per-keypoint visibility confidences

$$\mathbf{p}_t^{(v)} = \{(x_{t,k}^{(v)}, y_{t,k}^{(v)})\}_{k=1}^{K}, \qquad \mathbf{c}_t^{(v)} = \{c_{t,k}^{(v)}\}_{k=1}^{K}, \quad (1)$$

with $c_{t,k}^{(v)} \in [0, 1]$. The RL state concatenates per-view features,

$$\mathbf{s}_t = \left[\phi(\mathbf{p}_t^{(1)}, \mathbf{c}_t^{(1)}), \dots, \phi(\mathbf{p}_t^{(V)}, \mathbf{c}_t^{(V)})\right], \quad (2)$$

where ϕ normalizes pixel coordinates by image dimensions and applies confidence-based masking with hold-last imputation under occlusion.

Two properties of Equation (2) follow. First, $\mathbf{s}_t$ is defined entirely in image coordinates, so no extrinsics, depth, or object model is needed. Second, because ϕ acts per view and the features are concatenated rather than fused, a keypoint in view v is never geometrically associated with one in view v': no triangulation, no epipolar constraint, no shared metric frame. That association is what would require calibrated geometry; performing none of it is what makes the interface calibration-free. The same interface applies unchanged to deformable objects, where no pose or mesh model exists (Figure 3).

B. Final-State Relational Reward

A relation extractor $\mathcal{R}(\cdot)$ maps keypoints to a fixed-length vector of within-view 2D measurements—pairwise distances, signed coordinate differences encoding spatial ordering, and angular alignment terms: $\mathbf{g}_t^{(v)} = \mathcal{R}(\mathbf{p}_t^{(v)})$. For each task we annotate a target relation vector $\mathbf{g}_\star^{(v)}$ from a single final-state example per view, giving the per-view error $e_t^{(v)} = \left\| \mathbf{g}_t^{(v)} - \mathbf{g}_\star^{(v)} \right\|_2$ and view reliability $w_t^{(v)} = \frac{1}{K} \sum_k c_{t,k}^{(v)}$.

a) Occlusion safety: The natural confidence-scaled reward $r_t = -\sum_v w_t^{(v)} e_t^{(v)}$ is unsafe. As $w \to 0$ the reward tends to 0, which dominates every imperfectly solved state, so driving the scene into an untrackable configuration is rewarded. The crossover confidence at which occluding beats solving depends on the ratio between unsolved and solved error: $w = 0.075$ at a $\sim 13\times$ ratio but $w \approx 0.795$ at $1.25\times$, so the incentive is easiest

to reach on hard, high-residual-error tasks. With one of three views fully occluded, $w \approx 0.67$, already inside that regime. We therefore use a confidence-normalized reward with an explicit occlusion penalty:

$$r_t = -\frac{\sum_v w_t^{(v)} e_t^{(v)}}{\sum_v w_t^{(v)}} - \lambda \frac{1}{V} \sum_v \big(1 - w_t^{(v)}\big), \tag{3}$$

with an occlusion weight λ (0.35 in the controlled simulation of Section V-E). Normalization removes the $w \to 0$ optimum unconditionally rather than relying on the reachable set being constrained, and the penalty makes low confidence costly rather than free. Note that $r_t \leq 0$ by construction, with equality only at perfect completion under full confidence; every reported reward curve respects this bound. Success is defined by thresholding the terminal aggregated relation error,

$$\text{success} = \mathbb{1}\Big(\frac{\sum_v w_T^{(v)} e_T^{(v)}}{\sum_v w_T^{(v)}} < \tau\Big). \tag{4}$$

C. Belief-Gated Dual-Head Actor

Free-space and in-contact motion call for qualitatively different behavior, so we parameterize the actor with two Gaussian heads, $\pi_f = \mathcal{N}(\boldsymbol{\mu}_f(\mathbf{s}_t), \mathrm{diag}(\boldsymbol{\sigma}_f(\mathbf{s}_t)^2))$ and π_c defined analogously, and interpolate their parameters with a scalar gate $b_t \in [0, 1]$:

$$\begin{aligned} \boldsymbol{\mu}_t &= (1 - b_t)\boldsymbol{\mu}_f(\mathbf{s}_t) + b_t \boldsymbol{\mu}_c(\mathbf{s}_t), \\ \log \boldsymbol{\sigma}_t &= (1 - b_t) \log \boldsymbol{\sigma}_f(\mathbf{s}_t) + b_t \log \boldsymbol{\sigma}_c(\mathbf{s}_t), \end{aligned} \tag{5}$$

sampling $\mathbf{a}_t = \tanh(\boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \boldsymbol{\epsilon})$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$. Interpolating parameters rather than mixing densities keeps the policy a single tractable Gaussian, preserving SAC's closed-form log-likelihood.

The gate is produced by a two-state Bayesian filter over a latent mode $z_t \in \{F, C\}$, with $b_t \triangleq P(z_t = C \mid \mathcal{I}_{\leq t})$. Its observation y_t in Equation (6) is a thresholded reward change against an EMA baseline, and the prediction step is Equation (7):

$$y_t = \mathbb{1}\big(r_t - \bar{r}_{t-1} < -\delta\big), \qquad \bar{r}_t = (1 - \beta)\bar{r}_{t-1} + \beta r_t, \tag{6}$$

and the filter predicts and corrects in the standard way,

$$\hat{b}_t = b_{t-1}(1 - p_{\text{off}}) + (1 - b_{t-1})p_{\text{on}}, \tag{7}$$

$$b_t = \frac{P(y_t \mid C)\,\hat{b}_t}{P(y_t \mid C)\,\hat{b}_t + P(y_t \mid F)\,(1 - \hat{b}_t)}. \tag{8}$$

We use a relative threshold $\delta = 0.05$ after normalizing each task's initial geometric error to one, so this threshold needs no per-task tuning. The remaining filter constants—$p_{\text{on}} = 0.15$, $p_{\text{off}} = 0.10$, $P(y \mid C) = 0.75$, $P(y \mid F) = 0.20$, and $\beta = 0.1$—are the values used in the controlled simulation study of Section V-E.

a) Bayesian temporal inference: The transition probabilities in Equation (7) encode mode persistence; the likelihood ratio in Equation (8) updates that prior with current reward-change evidence. Unlike a fixed-weight EMA, the posterior response depends on both the prior belief and the observation likelihoods; the EMA in Equation (6) supplies an observation baseline, not the policy gate. Table VI tests whether simpler gates can recover the benefit of this inference and whether the gate actually identifies contact, and Section VI interprets the resulting control performance and its regime dependence.

The critics and entropy temperature follow standard SAC [20] with twin target critics; the only modification is the gated actor above.

IV. Experimental Setup

Because our claims are comparative, the protocol below fixes everything the compared methods share — hardware, sensing, control, reset, and interaction budget — so that each comparison isolates a single component. Results follow in Section V.

A. Platform

Experiments run on a 7-DoF Franka Research 3 with a parallel-jaw gripper and custom 3D-printed fingertips, under a low-level impedance controller at 1 kHz. Three RealSense cameras (D455 top-right, D435 side-top, D455 side-low) are mounted on an external frame and placed without any extrinsic calibration relative to the robot; only RGB streams are used and depth is discarded. The tracker and policy run on one workstation (RTX 6000, Core i9), with keypoints updating asynchronously at ≈45 Hz. The policy emits a 6-D Cartesian displacement $(\Delta x, \Delta y, \Delta z, \Delta\text{roll}, \Delta\text{pitch}, \Delta\text{yaw})$; rotational components are scaled for safe execution and impedance stiffness is fixed throughout training and evaluation.

B. Tasks and Keypoint Annotation

We evaluate seven tasks spanning rigid, articulated, and deformable interaction: fractured-piece alignment, broken-cup alignment, broken-bowl manipulation, towel folding, box opening, chain manipulation, and rope manipulation. Together they cover contact-guided pushing, frictional alignment, articulated constraints, and large-deformation dynamics. All seven tasks use the same three views. Annotation cost is identical in every case: selecting a handful of keypoints per view on the first frame, and labelling one final-state example from which $\mathbf{g}_\star$ is computed, paid once per task rather than per episode.

C. Protocol

Comparisons therefore isolate the representation (Section III-A), reward (Section III-B), and actor (Section III-C) rather than the learning machinery. All real-world hardware experiments, including baseline comparisons, ablations, contact-stability measurements, and tracking robustness evaluations, are evaluated across 3 random

seeds. We report the mean performance across these seeds; suite-level results additionally average over the seven tasks.

a) Interaction budget: Each RL method is trained for 20 episodes per task, each capped at 50 control steps and terminating early on success — at most 1000 environment steps, and fewer whenever episodes end early — taking 0.7–1.8 h of wall-clock time per task. All methods share this budget along with replay capacity and update-to-data ratio, so no method sees more interaction than another. Relation terms enter the reward unweighted, with per-view normalization balancing their scales, and keypoints below a tracking confidence of 0.45 are masked from both the state and the reward.

b) Baselines: The three RL baselines share our off-policy learner and each change exactly one component, so together they span our three claims. Single-Head SAC removes the belief gate and second head (isolating the gated actor); Vision-Based RL replaces the pixel-point state with a learned image encoder trained with augmentation [21] (isolating the representation); and Distance-Reward RL replaces the relational reward with a plain Euclidean distance to the goal keypoints (isolating the reward). Behavior Cloning trains a Diffusion Policy [22] on a fixed dataset of 50 teleoperated demonstrations per task and is the only method that does not interact online.

c) Train/test separation: Initial object configurations are drawn from disjoint training and evaluation sets: policies train on randomized configurations sampled within the workspace, and each task is then evaluated over 21 held-out randomized trials whose initial configurations were never seen during training. Episodes reset within a 50 cm × 30 cm workspace, with manual intervention only when an object leaves that region.

d) Metrics: Success thresholds the terminal aggregated relation error of Equation (4) at task-specific tolerances τ, fixed before evaluation: 0.03 for the alignment, folding, rope, and chain tasks, and 0.10 for box opening; Section V-B validates this criterion against physical success. Two metrics quantify behavior under contact. Instability Events counts, after first contact, each step at which the aggregated relation error increases beyond a fixed tolerance (mean per episode): how often a policy loses ground once engaged. Recovery Rate is the fraction of those events from which the error returns to or below its pre-event value within a fixed window: whether the policy re-establishes progress. Thresholds and windows are identical across methods.

V. Results

A. Real-Robot Comparison

Following the protocol of Section IV-C, Figure 4 and Table II report per-task success averaged over 3 random seeds. Our method averages 84.4 % across the suite, against 31.3 % for behavior cloning trained on 50 demonstrations per task, 35.4 % for vision-based RL, and 54.4 % for single-head SAC. The margin is largest on tasks whose progress depends on sustained frictional contact (chain and rope manipulation), where demonstrations cover the contact manifold sparsely and a policy must discover recovery behavior online. On towel folding the gap narrows, consistent with the task being solvable by a largely open-loop folding motion. Table I shows the same ordering in how policies behave once engaged: ours incurs the fewest instability events and recovers from the largest fraction of them.

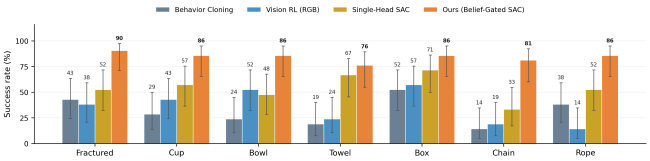


Fig. 4: Success rate on seven real-world tasks (21 held-out trials per task; means over 3 random seeds). The pixel-point interface with belief-gated SAC outperforms behavior cloning (50 demonstrations per task), vision-based RL, and single-head SAC across rigid, articulated, and deformable interaction.

TABLE I: Contact stability on the real robot (means over 3 random seeds); lower instability and higher recovery are better.

Method	Instab. events ↓	Recovery (%) ↑
Single-Head SAC	3.7	48.0
Distance Reward RL	3.1	52.0
Vision RL	4.4	39.0
Ours	0.9	78.0

B. Does Relational Success Reflect Physical Success?

Every success rate above uses the relational criterion of Equation (4), which thresholds tracked-keypoint relations rather than observing the physical outcome; a loose criterion could credit a policy that satisfies the pixel relations without completing the task. Table III therefore compares the relational label with physical task success determined independently of the tracked keypoints on the real-robot evaluation episodes. The two agree on 94.3 % of episodes with Cohen's $\kappa = 0.88$, far beyond chance agreement. Precision and recall are balanced (93.1 % and 95.0 %), so the criterion neither systematically over- nor under-credits outcomes; the few disagreements lean slightly toward crediting a success that did not physically occur. That residual is small against the 30-point margin over single-head SAC in Table II, so the reported ordering does not hinge on the proxy, and the same tracked relations can serve as both reward and evaluation criterion.

C. Ablations

Table IV ablates the pipeline. Removing belief gating costs the most on average (Section VI resolves this into a regime dependence). Substituting calibrated 3D keypoints for the pixel-point state lowers the average from 84.4 % to 70.1 %. View count matters monotonically:

TABLE II: Real-robot success (%), averaged over 3 random seeds, 21 trials per task; BC uses 50 demonstrations per task. Per-task means are rounded to the nearest percent.

Method	Frac.	Cup	Bowl	Towel	Box	Chain	Rope	Avg.
Behavior cloning	43	29	24	19	52	14	38	31.3
Vision RL (RGB)	38	43	52	24	57	19	14	35.4
Single-head SAC	52	57	48	67	71	33	52	54.4
Ours	90	86	86	76	86	81	86	84.4

TABLE III: Agreement between the relational success criterion of Equation (4) and physical task success on real-robot evaluation episodes. Physical success is the reference; the relational label is the prediction.

Accuracy	Precision	Recall	F1	Cohen's κ
94.3 %	93.1 %	95.0 %	94.0 %	0.88

averaged over the seven tasks, one camera reaches 69.4 %, two reach 80.3 %, and three reach 85.0 %. The largest single-view deficits are on fractured pieces and chain manipulation, where the gripper occludes the primary view — the same failure the multi-view aggregation of Equation (3) is designed to absorb.

D. Robustness of the Tracking Interface

Because the pixel-point state is the sole input to both policy and reward, tracker failure is the primary perceptual risk. We stress the interface under the two disturbances that dominate in practice: prolonged full occlusion of a view and degraded keypoint localization.

a) Recovery after full occlusion: We fully occlude one camera view and measure whether the tracker reacquires its points once the view is restored (Table V; 10 trials per task; means over 3 random seeds). Recovery is 100 % at every duration when the object is stationary, and remains at 92.9 % after 30 s when the object is displaced while hidden, which demands re-identification rather than persistence. Two properties produce this: TAPTRv3 predicts visibility explicitly, so an occluded point is masked rather than silently misplaced, and because relations are computed within each view, a lost view drops its own term from Equation (3) without corrupting the others.

b) Degraded keypoint localization: We inject perturbations at test time into an otherwise trained system, separating the two channels the keypoints feed. Jitter adds i.i.d. Gaussian noise $\mathcal{N}(0, 8^2)$ px to every keypoint of one randomly selected view; drift applies a zero-mean random walk to one view so that error accumulates. Corrupting the policy input leaves the reward clean and vice versa. Degradation is gradual rather than catastrophic (Table V), with two asymmetries: drift costs more than jitter, because accumulated displacement biases the relations while zero-mean noise averages out, and corrupting the reward costs more than the state, because an erroneous reward misdirects learning rather than adding recoverable observation noise. Confidence masking and multi-view aggregation bound these costs, since unreliable points are down-weighted and uncorrupted views still supply stable feedback.

TABLE IV: Real-robot ablations, averaged over 3 random seeds and seven tasks (success rate %, 21 trials per task).

Variant	Avg. success (%)
Full model (pixel-point, belief-gated)	84.4
w/o belief gating (single-head)	54.4
w/o relational reward	62.6
Single view only	69.4
w/o impedance control	58.5
3D keypoints instead of pixel	70.1

TABLE V: Robustness of the tracking interface (means over 3 random seeds). Top: recovery (%) after full occlusion of one view (10 trials per task); "moved" displaces the object while hidden. Bottom: success-rate drop (%) under injected keypoint corruption, by corrupted channel.

Occlusion	3 s	10 s	30 s
Object static	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Object moved	100.0 ± 0.0	95.7 ± 2.4	92.9 ± 3.1
Corruption	**Policy in**	**Reward in**	**Drop (%)**
Clean	clean	clean	0.0
State jitter	corrupt	clean	$2.3\text{–}6.3 \pm 0.6$
Reward jitter	clean	corrupt	$3.5\text{–}9.4 \pm 0.9$
State drift	corrupt	clean	$5.1\text{–}8.9 \pm 1.3$
Reward drift	clean	corrupt	$6.5\text{–}10.1 \pm 1.6$

E. Mechanism Studies in Controlled Simulation

The remaining questions—whether the gate's benefit comes from Bayesian inference or added capacity, whether the gate identifies contact, whether the reward correction costs anything, and whether the system tolerates camera displacement—require ground truth and interventions the hardware cannot provide. We study them in a MuJoCo environment reproducing the paper's interface: multi-view keypoints with per-point confidence, a pixel-space relational reward, and contact-rich planar alignment, with ground-truth contact, controllable occlusion, and settable camera pose. These are studies of mechanism, not reproductions of the hardware results; absolute values are not comparable to the real-robot numbers above. Unless noted, n denotes seeds and intervals are standard errors.

a) Is Bayesian temporal inference essential?: Table VI compares eight actor variants over 10 seeds each. Increasing single-head capacity changes success from 40.3 % to 41.1 %, a fixed gate reaches 42.8 %, random smooth and EMA gates 41.9 % and 47.2 %, and a learned gate 44.5 %. Bayesian belief gating reaches 60.8 %: about 20 points over the capacity-matched single head, 14 over EMA, and 16 over the learned gate, with a 7-point gap to the contact oracle; its standard-error interval overlaps no other non-oracle variant. It is also the only non-oracle

TABLE VI: Targeted gate ablation in simulation (mean ± s.e. over 10 seeds). Success in %. Contact F1 and onset delay (control steps from physical contact to the gate switching on) are scored against ground-truth contact; single-head variants have no gate. Coll.: seeds (of 10) whose training collapsed. The contact oracle uses privileged contact information.

Method	Success ↑	F1 ↑	Delay ↓	Coll. ↓
Single-head SAC	40.3 ± 3.8	—	—	2/10
Cap.-matched single	41.1 ± 4.1	—	—	2/10
Fixed gate	42.8 ± 3.5	0.65 ± 0.03	0.9 ± 0.2	2/10
Random smooth gate	41.9 ± 4.4	0.61 ± 0.05	2.1 ± 0.4	3/10
EMA gate	47.2 ± 3.6	0.73 ± 0.02	2.5 ± 0.3	1/10
Learned gate	44.5 ± 4.0	0.75 ± 0.03	1.8 ± 0.2	2/10
Bayesian gate (ours)	$\mathbf{60.8 \pm 3.1}$	$\mathbf{0.87 \pm 0.02}$	1.4 ± 0.2	0/10
Oracle contact	68.1 ± 2.5	1.00	0	0/10

variant with no collapsed seed (0/10, against 1–3/10 for the others). Neither extra capacity, fixed blending, nor generic smoothing recovers its performance, which supports Bayesian temporal inference as an essential component in this setting.

b) Does the Bayesian gate identify contact?: Higher success alone does not show that the gate improves control by inferring contact. The contact columns of Table VI test this against the simulator's ground-truth contact signal, scoring each gate's binary output by per-step contact F1 and by onset delay, the number of control steps between physical contact and the gate switching on. The Bayesian gate reaches F1 0.87, against 0.61–0.75 for the fixed, random smooth, EMA, and learned gates, and 0.68 for using the thresholded reward-change observation y_t of Equation (6) directly as the gate (not tabulated: 0.8-step delay, 43 % success). The simpler gates expose the trade-off the filter resolves: the instantaneous threshold responds fastest but is least accurate, because single-step reward changes are noisy, while the EMA gate suppresses that noise only by lagging (2.5 steps). The Bayesian filter is most accurate at an intermediate delay (1.4 steps): the transition prior of Equation (7) rejects isolated false observations without a fixed lag. Control success tracks contact F1. Every gate at or below F1 0.75 stays within 42–47 %, the Bayesian gate's 0.87 yields 60.8 %, and the oracle (F1 1.00) reaches 68.1 %; the 7-point gap to the oracle bounds what better contact inference could still recover.

c) Regime dependence: Table VII reports a separate architecture study under identical state, reward, and budget within each regime ($n = 8$). Belief gating outperforms the learned gate on the transient-contact task; the sustained-contact column bounds where the mechanism applies.

On a stick-slip task, belief gating falls below single-head SAC (0.54 ± 0.11 versus 0.67 ± 0.05), collapsing in 3 of 8 seeds where single-head collapses in none, whereas under the planar alignment of Table VI it collapses in no seed. This limits the scope of the benefit established there (Section VI).

TABLE VII: Separate contact-regime study in simulation (success fraction, mean ± s.e.). These estimates are distinct from the targeted gate ablation in Table VI and the real-robot results.

Actor	Heads/gate	Transient	Sustained
Single-head SAC	1 / none	0.37 ± 0.09	$\mathbf{0.67 \pm 0.05}$
Ensemble-uncert.	1 / uncert.	0.39 ± 0.19	—
Learned-gate SAC	2 / learned	0.42 ± 0.07	0.55 ± 0.08
Belief-gated (ours)	2 / belief	$\mathbf{0.59 \pm 0.06}$	0.54 ± 0.11
Oracle-contact SAC	2 / oracle	0.71 ± 0.04	0.53 ± 0.09

d) Reward correction is free: Comparing the three reward forms under learning, the confidence-normalized reward of Equation (3) attains 0.63 ± 0.12 success against 0.56 ± 0.18 for an unweighted reward and 0.37 ± 0.19 for the original confidence-scaled form, with correspondingly lower terminal error; removing the degenerate optimum costs nothing.

e) Camera placement: Retraining from scratch under randomly perturbed camera rigs gives 41.3, 61.3, 48.0, and 52.0 % success at 0, 5, 10, and 20 cm of camera displacement ($n = 3$ each): no degradation across the range. This establishes calibration-free deployment (cameras placed anew, then trained without measuring extrinsics), as distinct from placement invariance, a trained policy surviving camera motion, which would require zero-shot transfer across rigs and lies outside this evaluation.

f) Demonstration budget: Behavior cloning on the same interface scales smoothly with demonstrations: 44.0, 34.7, 50.7, 69.3, 77.3, and 88.0 % at 5, 10, 25, 50, 100, and 200 demonstrations. Imitation therefore solves the simulated task given enough coverage. This simulation-only sweep is separate from the real-world BC baseline (50 demonstrations per task); environment steps and demonstrations are not commensurable units, and the budgets were not matched.

VI. Analysis, Limitations, and Conclusion

a) The role of Bayesian temporal inference: The controls in Table VI show that capacity and smoothness alone do not recover the Bayesian gate's performance, and its contact columns supply the mechanism: the gate identifies ground-truth contact more accurately than the fixed, smooth, EMA, learned, and instantaneous alternatives (F1 0.87 versus 0.61–0.75), control success tracks that accuracy, and the contact oracle bounds what perfect identification would add. The gate helps because it infers the contact mode, not because it smooths; the comparison does not separate the transition prior from the observation model. The regime study in Table VII bounds this benefit: belief gating outperforms single-head SAC under transient contact but underperforms it under sustained stick-slip. The contact-identification study was not repeated under sustained contact, so whether that regression reflects degraded contact inference or a limitation of the gated policy itself remains untested.

b) Limitations: The gate's benefit is regime-dependent: it underperforms single-head SAC under sustained stick-slip. Addressing this regression remains open. The relational reward is restricted to static geometric goals; periodic motion requires a phase-indexed target to avoid rewarding a stationary configuration. Keypoints and target relations are annotated manually once per task; automating this remains open. Finally, confidence masking and multi-view aggregation mitigate but do not eliminate tracker failures, especially accumulating drift (Section V-D).

c) Broader impact: Lowering the barrier to real-world RL makes deployment accessible to operators with limited robotics expertise; online exploration requires workspace limits and impedance bounds, and RGB recordings of shared workspaces require privacy protections.

d) Conclusion: We presented a calibration-free pixel-point interface whose tracked keypoints define both state and relational reward. It outperforms behavior cloning, vision-based RL, and single-head SAC on seven real-world tasks, with hardware results averaged over 3 random seeds and BC trained on 50 demonstrations per task, and its relational success criterion agrees with physical success on 94.3 % of evaluation episodes ($\kappa = 0.88$). Targeted controls support Bayesian temporal inference as an essential algorithmic component: belief gating achieves 60.8 % success over 10 seeds with no collapsed seed, exceeding capacity-matched, fixed, smooth, and learned alternatives, and does so by identifying contact (F1 0.87 against ground truth, versus at most 0.75 for simpler gates). The separate regime study establishes a sustained-contact limitation that motivates further investigation.

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